

**MAHATMA EDUCATION SOCIETY'S
PILLAI COLLEGE OF ARTS, COMMERCE & SCIENCE
(Autonomous)
NEW PANVEL**

**PROJECT REPORT ON
“Student Record System”
IN PARTIAL FULFILLMENT OF
BACHELOR OF COMPUTER SCIENCE
SEMESTER VI – 2024-25**

**PROJECT GUIDE
Prof. Jabeen Masum Ma'am**

SUBMITTED BY: Lakshmi Prasanth Pillai

ROLL NO: 9165

Evaluation sheet for continuous assessment with rubrics

Class: BSc Computer Science

Subject: Software testing and quality assurance

Name of Evaluator: Jabeen Masum Ma'am

Details about the continuous Assessment 2/Project work

Title: "Student Record System "



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Name of the Student : Lakshmi Prasanth Pillai

Roll Number : 9165

Class / Division : TYCS A

Please circle appropriate score

Grading Criteria	Fair	Good	Excellent	Total
Introduction/ Description of the Case	1	2	3	/3
SWOT Analysis of the company used for case analysis pertaining to the case (strength of CA2 topic: e.g main important feature of CA1, Weakness : limitations of the project, Opportunities : in carrier in the future, Threat : obstacles that can cause failure to project CA2	3	4	5	/5
Learnings from the case	2	3	4	/4
Delivery/presentation skills	1	2	3	/3
Total				/15

- Inform the class the rubric format and the method of evaluation.

Co-ordinator,
 Shubhangi Pawar

INTRODUCTION:

In the realm of software development, ensuring the quality and reliability of applications like the Student Record System is paramount. A well-defined testing strategy plays a crucial role in achieving this objective by systematically identifying and addressing potential issues. This document outlines the testing methodologies employed to validate the functionality, performance, and robustness of the Student Record System, leveraging both manual and automated approaches. The goal is to ensure a seamless and reliable experience for administrators, students, and other stakeholders.

Testing Methodologies Utilized

validate the functionality, performance, and robustness of the application, the following testing methodologies are employed:

1. Manual Testing: Excel and Selenium IDE

The testing process commenced with manual testing facilitated through **Excel spreadsheets**. Test cases were meticulously crafted to assess critical functionalities such as login, add a course, add a new student, edit a student, change password. This approach allows for structured documentation and systematic execution of test scenarios.

While **Selenium IDE** is a powerful tool for automated testing, in this project, manual testing was also performed using selenium ide. The decision to opt for manual testing was made to ensure meticulous attention to detail and thorough validation of each step in the process.

1. Updating admin profile
2. Edit a new course
3. Delete a course
4. Add a new subject
5. Password Recovery

2. Automated Testing: Eclipse IDE

Further automation was achieved through the utilization of Eclipse IDE, enabling the creation and execution of test scripts using Java tailored specifically for the Microsoft Edge browser. Test scenarios covering Logout, edit a subject, update a session, delete a student were automated to enhance test coverage and expedite testing cycles.

SWOT Analysis for Testing the Student Record System

- **Strengths**

Comprehensive Manual Testing: Detailed manual test cases designed in Excel ensure thorough validation of critical functionalities like login, student management, and password recovery.

Tool Versatility: Combination of manual (Excel, Selenium IDE) and automated testing (Eclipse IDE) allows for a balanced testing approach. Tailored test scenarios for Microsoft Edge browser ensure compatibility across platforms.

- **Weakness**

Tool Dependency: Heavy reliance on Selenium IDE for manual testing and Eclipse IDE for automation may create bottlenecks if tools are unavailable or incompatible.

Time-Consuming Manual Testing: Manual testing, while thorough, can be slower compared to fully automated approaches for larger or more complex test scenarios.

- **Opportunities**

Enhanced Reporting: Adopting advanced reporting tools or integrating with CI/CD pipelines could streamline result analysis and improve team collaboration.

Integration with Advanced Tools: Incorporating tools like Selenium WebDriver or TestNG could improve automation efficiency and scalability. Utilizing cloud-based testing platforms could enhance cross-browser and cross-device testing capabilities.

- **Threats**

Evolving Requirements: Frequent changes in system requirements or features could impact the stability of test scripts and manual test cases.

Tool Limitations: Dependence on specific tools like Selenium IDE may pose challenges if features are deprecated or unsupported in future updates.

Manual testing (EXCEL SHEET)

LOGIN

Application Name	Student Record System				
Scenario	Login Page				
Tester Name	Lakshmi Pillai				
Developer Name	Lakshmi Pillai				

Sr.no	Scenario test case/name	Steps	Input	Expected Output	Current Output	Status
1	Login with valid username and password	Start browser Open url Open login page Enter Username Enter password Click on submit	Microsoft edge http://localhost/studentrecordms/login.php Login.php admin Test@12345	Login successful	Login successful	Pass
2	Login with invalid username and valid password	Start browser Open url Open login page Enter username Enter password Click on submit	Microsoft edge http://localhost/studentrecordms/login.php Login.php Student Test@12345	Login unsuccessful	Login unsuccessful	
3	Login with valid username and invalid password	Start browser Open url Open login page Enter username Enter password Click on submit	Microsoft edge http://localhost/studentrecordms/login.php Login.php admin Password9165	Login unsuccessful	Login unsuccessful	
4	Login with invalid username and invalid password	Start browser Open url Open login page Enter username Enter password Click on submit	Microsoft edge http://localhost/studentrecordms/login.php Login.php lakshmi 9165abc	Login unsuccessful	Login unsuccessful	
5	Login with blank username and blank password	Start browser Open url Open login page Enter username Enter password Click on submit	Microsoft edge http://localhost/studentrecordms/login.php Login.php	Login unsuccessful	Login unsuccessful	

Add a new student

Application Name	Student Record System				
Scenario	Add a new student				
Tester Name	Lakshmi Pillai				
Developer Name	Lakshmi Pillai				

Sr.no	Scenario test case/name	Steps	Input	Expected Output	Current Output	Status
1	enter all valid data	Start browser open url open register page register(select course) select subject Personal Informations(First name) Family income Gender Category Physically challenged Nationality Contact information(mobile no) email id country state city address Academic Informations (board) roll no pass year subject marks obtained full marks Click on Register	Microsoft edge http://localhost:studentrecords/dashboard.php register.php BA English+Sociology+ Philosophy Prasant 700000 Male general No Indian 2213243656 prasant@yahoo.com india Delhi New Delhi anmol society MUMBAI UNIVERSITY 9165 2023 2024 Software testing and quality assurance 60	Student Registration successful	Student Registration successful	pass

2 enter alphabets in palce of numbers	Start browser	Microsoft edge	Student Registration unsuccessful	Student Registration unsuccessful	pass
	open url	http://localhost/studentrecords/dashboard.php			
	open register page	register.php			
	register(select course)	BA			
	select subject	English+Sociology+ Philosophy			
	Personal Informations(First name)	Prasant			
	Family income	700000			
	Gender	Male			
	Category	general			
	Physically challenged	No			
	Nationality	Indian			
	Contact Information(mobile no)	esffcsddff			
	email id	prasant@yahoo.com			
	country	india			
	state	Delhi			
	city	New Delhi			
	address	anmol society			
	Academic Informations (board)	MUMBAI UNIVERSITY			
	roll no	9165			
	pass year	2023 2024			
	subject	Software testing and quality assurance			
	marks obtained full marks	60			
	Click on Register				
3 not selecting any gender income and category	Start browser	Microsoft edge	Student Registration unsuccessful	Student Registration unsuccessful	pass
	open url	http://localhost/studentrecords/dashboard.php			
	open register page	register.php			
	register(select course)	BA			
	select subject	English+Sociology+ Philosophy			
	Personal Informations(First name)	Prasant			
	Family income				
	Gender				
	Category				
	Physically challenged	No			
	Nationality	Indian			
	Contact Information(mobile no)	esffcsddff			
	email id	prasant@yahoo.com			
	country	india			
	state	Delhi			
	city	New Delhi			
	address	anmol society			
	Academic Informations (board)	MUMBAI UNIVERSITY			
	roll no	9165			
	pass year	2023 2024			
	subject	Software testing and quality assurance			
	marks obtained full marks	60			
	Click on Register				
4 enter all blank data	Start browser	Microsoft edge	Student Registration unsuccessful	Student Registration unsuccessful	pass
	open url	http://localhost/studentrecords/dashboard.php			
	open register page	register.php			
	register(select course)				
	select subject				
	Personal Informations(First name)				
	Family income				
	Gender				
	Category				
	Physically challenged				
	Nationality				
	Contact Information(mobile no)				
	email id				
	country				
	state				
	city				
	address				
	Academic Informations (board)				
	roll no				
	pass year				
	subject				
	marks obtained full marks				
	Click on Register				

Add a new course

Application Name	Student Record System		
Scenario	Add a new course		
Tester Name	Lakshmi Pillai		
Developer Name	Lakshmi Pillai		

Sr.no	Scenario test case/name	Steps	Input	Expected Output	Current Output	Status
1	enter all valid data	Start browser open url open add course page enter course short name enter course Full name click create course	Microsoft edge http://localhost/studentrecords/dashboard.php add-course.php MSc IT MASTERS IN INFORMATION TECHNOLOGY	Course Added successfully	Course Added successfully	Pass
2	Add blank short name and enter valid full name	Start browser open url open add course page enter course short name enter course Full name click create course	Microsoft edge http://localhost/studentrecords/dashboard.php add-course.php MASTERS IN INFORMATION TECHNOLOGY	Course Added unsuccessful	Course Added unsuccessful	Pass
3	Add short name and blank full name	Start browser open url open add course page enter course short name enter course Full name click create course	Microsoft edge http://localhost/studentrecords/dashboard.php add-course.php MSc IT	Course Added unsuccessful	Course Added unsuccessful	Pass
4	Add blank short name and full name	Start browser open url open add course page enter course short name enter course Full name click create course	Microsoft edge http://localhost/studentrecords/dashboard.php add-course.php	Course Added unsuccessful	Course Added unsuccessful	Pass

Change-password

Application Name	Student Record System		
Scenario	change-password		
Tester Name	Lakshmi Pillai		
Developer Name	Lakshmi Pillai		

Sr.no	Scenario test case/name	Steps	Input	Expected Output	Current Output	Status
1	enter all valid current password and confirm password	Start browser Open url Open change-password page Enter Current Password Enter New Password Enter Confirm Password Click change	Microsoft edge http://localhost/studentrecords/change-password.php change-password.php test@12345 Test@12345 Test@12345	password successfully changed.	password successfully changed.	Pass
2	enter invalid current password and valid new, confirm password	Start browser Open url Open change-password page Enter Current Password Enter New Password Enter Confirm Password Click change	Microsoft edge http://localhost/studentrecords/change-password.php change-password.php student Test@12345 Test@12345	password changed unsuccessful.	password changed unsuccessful.	Pass
3	enter valid current password and invalid new, current password	Start browser Open url Open change-password page Enter Current Password Enter New Password Enter Confirm Password Click change	Microsoft edge http://localhost/studentrecords/change-password.php change-password.php test@12345 Password9165 Password9165	password changed unsuccessful.	password changed unsuccessful.	Pass
4	enter invalid current, new confirm password	Start browser Open url Open change-password page Enter Current Password Enter New Password Enter Confirm Password Click change	Microsoft edge http://localhost/studentrecords/change-password.php change-password.php admin Password9165 Password9165	password changed unsuccessful.	password changed unsuccessful.	Pass
5	Login with blank username and blank password	Start browser Open url Open change-password page Enter Current Password Enter New Password Enter Confirm Password Click change	Microsoft edge http://localhost/studentrecords/change-password.php change-password.php	password changed unsuccessful.	password changed unsuccessful.	Pass

Edit Student

Application Name	Student Record System				
Scenario	Edit a student				
Tester Name	Lakshmi Pillai				
Developer Name	Lakshmi Pillai				

Sr.no	Scenario test case/name	Steps	Input	Expected Output	Current Output	Status
1	updating all valid previous data	Start browser open url open register page register(select course) select subject Personal Informations(First name) Family income Gender Category Physically challenged Nationality Contact Information(mobile no) email id country state city address Academic Informations (board) roll no pass year subject marks obtained full marks Click on Update	Microsoft edge http://localhost/studentrecordsms/dashboard.php edit-student.php BA English+Sociology+ Philosophy Prasant 700000 Male general No Indian 2213243656 prasant@yahoo.com india Delhi New Delhi anmol society MUMBAI UNIVERSITY 9165 2023 2024 Software testing and quality assurance 60	Update successful	Update successful	pass
2	blank values in palce of mobile no	Start browser open url open register page register(select course) select subject Personal Informations(First name) Family income Gender Category Physically challenged Nationality Contact Information(mobile no) email id country state city address Academic Informations (board) roll no pass year subject marks obtained full marks Click on Update	Microsoft edge http://localhost/studentrecordsms/dashboard.php edit-student.php BA English+Sociology+ Philosophy Prasant 700000 Male general No Indian prasant@yahoo.com india Delhi New Delhi anmol society MUMBAI UNIVERSITY 9165 2023 2024 Software testing and quality assurance 60	update unsuccessful	Update unsuccessful	pass
3	updating all needed valid data	Start browser open url open register page register(select course) select subject Personal Informations(First name) Family income Gender Category Physically challenged Nationality Contact Information(mobile no) email id country state city address Academic Informations (board) roll no pass year subject marks obtained full marks Click on Update	Microsoft edge http://localhost/studentrecordsms/dashboard.php edit-student.php BA English+Sociology+ Philosophy Prasant 700000 Male general No Indian 2213243656 prasant@yahoo.com india Delhi New Delhi anmol society MUMBAI UNIVERSITY 9165 2023 2024 Software testing and quality assurance 100	Update successful	Update successful	pass

Manual testing (SELENIUM)

- ▼ DELETE A COURSE*
 - ✓ delete a new course*
 - ▼ add a subject*
 - ✓ add a subject*
 - ✓ add a subject (blank)*
 - ✓ add a subject (no course)*
 - ▼ change email id in admin profile*
 - ✓ Change admin email id (blank)*
 - ✓ Change admin email id*
 - ✓ Change admin name*
 - ✓ Change admin name (blank)*
 - ▼ edit/delete a new course*
 - ✓ edit new course*
 - ✓ delete a new course*
 - ▼ password recovery*
 - ✓ PASSWORD RECOVERY*
 - ✓ blank id and email pass recovery*
 - ✓ invalid id and email pass recovery*
-

● UPDATING ADMIN PROFILE

▼ change email id in admin pr...

✓ Change admin email id (blank)*

✓ Change admin email id*

✓ Change admin name*

✓ Change admin name (blank)*

Updating admin name :

1. Leaving name blank

WELCOME ADMIN

Edit Admin Profile

Name

Email id

Admin12345

 Please fill out this field.

Logind Id/ username

admin

Logind Id/ username can't be edit

Update Profile

Command	Target	V
✓ open	http://localhost/studentrecordms/admin-profile.php	
✓ set window size	1050x748	
✓ click	css=.form-group:nth-child(1)	
✓ click	id=adminname	
✓ type	id=adminname	
✓ click	name=submit	

2. Entering valid name :

WELCOME ADMIN

Edit Admin Profile

Name

Admin

Email id

Admin12345@gmail.com

Logind Id/ username

admin

Logind Id/ username can't be edit

Update Profile

1	✓ open	http://localhost/studentrecordms/admin-profile.php	
2	✓ set window size	1050x748	
3	✓ click	id=adminname	
4	✓ type	id=adminname	Admin
5	✓ click	name=submit	

Updating admin email id

1. Leaving email id space blank

✓ open	http://localhost/studentrecordms/admin-profile.php
✓ set window size	1050x748
✓ click	id=aemailid
✓ type	id=aemailid
✓ click	name=submit

WELCOME ADMIN

Edit Admin Profile

Name

Email id

Logind Id/ username

Logind Id/ username can't be edit

Please fill out this field.

2. Enter valid email id

in current test Ctrl+R records/admin-profile.php			
	Command	Target	Value
1	✓ open	http://localhost/studentrecords/admin-profile.php	
2	✓ set window size	1050x748	
3	✓ click	id=aemailid	
4	✓ click	id=aemailid	
5	✓ click	id=aemailid	
6	✓ click	id=aemailid	
7	✓ type	id=aemailid	Admin12345@gmail.com
8	✓ click	name=submit	

WELCOME ADMIN

Edit Admin Profile

Name

Email id

Logind Id/ username

Logind Id/ username can't be edit

Update Profile

EDIT NEW COURSE

http://localhost Test execution speed ge-courses.php			
Command	Target	Value	
1 ✓ open	http://localhost/studentrecords/manage-courses.php		
2 ✓ set window size	1050x748		
3 ✓ click	css=.gradeX:nth-child(10) .btn-primary		
4 ✓ click	id=cfull		
5 ✓ type	id=cfull	BACHELOR OF INFORMATION TECHNOLOGY	
6 ✓ click	name=submit		

S No ^	Short Name ^	Full Name ^	Created Date ^	Action ^
1	MCA	MASTER OF COMPUTER APPLICATION	15-02-2024	Edit Delete
2	B.TECH	BACHELOR OF TECHNOLOGY	15-02-2024	Edit Delete
3	BCA	BACHELOR OF COMPUTER APPLICATION	15-02-2024	Edit Delete
4	BA	BACHELOR OF ARTS	15-02-2024	Edit Delete
5	MA	MASTER OF ARTS	15-02-2024	Edit Delete
6	BSC CS	BACHELOR OF COMPUTER SCIENCE	26-12-2024	Edit Delete
7	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete
8	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete
9	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete
10	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete

Showing 1 to 10 of 10 entries

[Previous](#) [1](#) [Next](#)

			cid	cshort	cfull	cdate	update_date
☐	Edit	Copy	Delete	1	MCA	MASTER OF COMPUTER APPLICATION	15-02-2024 NULL
☐	Edit	Copy	Delete	3	B.Tech	Bachelor of Technology	15-02-2024 NULL
☐	Edit	Copy	Delete	4	BCA	Bachelor of Computer Application	15-02-2024 NULL
☐	Edit	Copy	Delete	5	BA	Bachelor of Arts	15-02-2024 NULL
☐	Edit	Copy	Delete	9	MA	Master of Arts	15-02-2024 NULL
☐	Edit	Copy	Delete	10	BSc CS	BACHELOR OF COMPUTER SCIENCE	26-12-2024 NULL
☐	Edit	Copy	Delete	12	BSc IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024 NULL
☐	Edit	Copy	Delete	13	BSc IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024 NULL
☐	Edit	Copy	Delete	14	BSc IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024 NULL
☐	Edit	Copy	Delete	15	BSc IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024 NULL

● ADD A NEW SUBJECT

▼ add a subject*

✓ add a subject*

✓ add a subject (blank)*

✓ add a subject (no course)*

1. Enter all valid data

✓ click	link text=Add subject					
✓ click	id=cshort					
✓ click	id=cshort					
✓ click	id=cshort					
✓ select	id=cshort					label=BSc CS (BACHELOR OF COMPUTER SCIENCE)
✓ click	name=sub1					
✓ type	name=sub1					Java
✓ click	name=sub2					
✓ type	name=sub2					web service
✓ click	name=sub3					
✓ type	name=sub3					C++
✓ click	name=sub4					
✓ type	name=sub4					C
✓ click	name=submit					

6	BACHELOR OF COMPUTER SCIENCE	JAVA	WEB SERVICE	C++	C	Edit Delete
7	MASTER OF COMPUTER APPLICATION	SOFTWARE TESTING AND QUALITY ASSURANCE	CLOUD COMPUTING	MACHINE LEARNING	ANDROID	Edit Delete
8	BACHELOR OF COMPUTER SCIENCE	JAVA	WEB SERVICE	C++	C	Edit Delete
9	BACHELOR OF COMPUTER SCIENCE	JAVA	WEB SERVICE	C++	C	Edit Delete

☐	Edit Copy Delete	7 10	BACHELOR OF COMPUTER SCIENCE	Java	web service	C++	C	2024-12-27 22:31:39	NULL
☐	Edit Copy Delete	9 1	MASTER OF COMPUTER APPLICATION	Software testing and quality assurance	Cloud Computing	Machine Learning	Android	2024-12-27 23:19:10	NULL
☐	Edit Copy Delete	10 10	BACHELOR OF COMPUTER SCIENCE	Java	web service	C++	C	2024-12-27 23:23:58	NULL
☐	Edit Copy Delete	13 10	BACHELOR OF COMPUTER SCIENCE	Java	web service	C++	C	2024-12-29 02:37:22	NULL
☐	Edit Copy Delete	14 10	BACHELOR OF COMPUTER SCIENCE	Java	web service	C++	C	2024-12-29 04:39:06	NULL

2. Not entering any data

✓ open	http://localhost/studentrecordms/add-subject.php	
✓ set window size	1050x748	
✓ click	id=cshort	
✓ select	id=cshort	label=BSc CS (BACHELOR OF COMPUTER SCIENCE)
✓ click	name=submit	

Subject 1

 Please fill out this field.

Subject 2

Subject 3

Subject 4

3. Not choosing course and entering data

✓ open	http://localhost/studentrecordms/add-subject.php	
✓ set window size	1050x748	
✓ click	name=sub1	
✓ type	name=sub1	Java
✓ click	name=sub2	
✓ type	name=sub2	web service
✓ click	name=sub3	
✓ type	name=sub3	C++
✓ click	name=sub4	
✓ type	name=sub4	C
✓ click	name=submit	

Course Short Name*

SELECT 

 Please select an item in the list.

Subject 1

Java

Subject 2

web service

● DELETE A COURSE

✓ delete a new course*	http://localhost/studentrecordms/manage-courses.php		
	Command	Target	Value
1	✓ open	http://localhost/studentrecordms/manage-courses.php	
2	✓ set window size	1050x748	
3	✓ choose ok on next confirmation		
4	✓ click	css=gradeX:nth-child(10) .btn-danger	
5	✓ choose ok on next confirmation	Do you really wan to delete?	
6	✓ webdriver choose ok on visible confirmation		

Show 10 entries Search:

S No	Short Name	Full Name	Created Date	Action
1	MCA	MASTER OF COMPUTER APPLICATION	15-02-2024	Edit Delete
2	B.TECH	BACHELOR OF TECHNOLOGY	15-02-2024	Edit Delete
3	BCA	BACHELOR OF COMPUTER APPLICATION	15-02-2024	Edit Delete
4	BA	BACHELOR OF ARTS	15-02-2024	Edit Delete
5	MA	MASTER OF ARTS	15-02-2024	Edit Delete
6	BSC CS	BACHELOR OF COMPUTER SCIENCE	26-12-2024	Edit Delete
7	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete
8	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete
9	BSC IT	BACHELOR OF INFORMATION TECHNOLOGY	27-12-2024	Edit Delete

● PASSWORD RECOVERY

▼ password recovery*

✓ PASSWORD RECOVERY*

✓ blank id and email pass recovery*

✓ invalid id and email pass recovery*

✓ invalid id pass recovery*

1. Enter all valid data and new password

http://localhost/studentrecordsms/login.php			
	Command	Target	Value
1	✓ open	http://localhost/studentrecordsms/login.php	
2	✓ set window size	1050x748	
3	✓ click	linkText=Password Recovery	
4	✓ type	id=id	admin
5	✓ click	id=emailid	
6	✓ type	id=emailid	Admin12345@
7	✓ click	id=emailid	
8	✓ type	id=emailid	Admin12345@gmail.com
9	✓ click	id=password	
10	✓ type	id=password	test@12345
11	✓ click	name=submit	

Sign In

[Login](#)

2. Enter invalid id and valid email

1	✓ open	http://localhost/studentrecordsms/login.php	
2	✓ set window size	1050x748	
3	✓ click	linkText=Password Recovery	
4	✓ type	id=id	laxmi
5	✓ click	id=emailid	
6	✓ type	id=emailid	Admin12345@
7	✓ click	id=emailid	
8	✓ click	id=emailid	
9	✓ type	id=emailid	Admin12345@gmail.com
10	✓ click	id=password	
11	✓ type	id=password	test@12345
12	✓ click	name=submit	

Sign In

laxmi

Admin12345@gmail.com

.....|

Login

Change Password

localhost says

invalid details

OK

3. Enter invalid id and invalid email

✓ open	http://localhost/studentrecordsms/login.php	
✓ set window size	1050x748	
✓ click	linkText=Password Recovery	
✓ type	id=id	laxmi
✓ click	id=emailid	
✓ type	id=emailid	12345@
✓ click	id=emailid	
✓ type	id=emailid	12345@gmail.com
✓ click	id=password	
✓ type	id=password	test@12345
✓ click	name=submit	

Sign In

Login

Change Password



4. Enter blank id and email

✓ open	http://localhost/studentrecordms/login.php	
✓ set window size	1050x748	
✓ click	linkText=Password Recovery	
✓ click	id=password	
✓ type	id=password	test@12345
✓ click	name=submit	

Sign In

Ad

! Please fill out this field.

Login

Change Password

AUTOMATED TESTING

1. Automated testing [Logout] Eclipse :

```

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.By;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.JavascriptExecutor;
import java.time.Duration;
public class LogoutAutomationTest {
    public static void main(String[] args) {
        // Set path to your EdgeDriver executable
        // System.setProperty("webdriver.edge.driver", "path_to_your_edge_driver");
        // Initialize the EdgeDriver
        WebDriver driver = new EdgeDriver();
        try {
            // Open the website
            driver.get("http://localhost/studentrecordms/dashboard.php");
            // Wait for the Logout link to be present and clickable
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(60)); // Increased
timeout
            WebElement logoutLink =
wait.until(ExpectedConditions.elementToBeClickable(By.xpath("//a[contains(., 'Logout')]")));
            // Optionally, use JavaScript to click the element if it's not interactable
            JavascriptExecutor js = (JavascriptExecutor) driver;
            js.executeScript("arguments[0].click();", logoutLink);
            // Wait for the logout to complete (optional)
            Thread.sleep(2000); // Pause for 2 seconds
            System.out.println("Successfully logged out.");
        } catch (Exception e) {
            e.printStackTrace();
        } finally {
            // Close the browser
            driver.quit();
        }
    }
}

```

Student Record Management System

Sign In

[Password Recovery](#)

login

```
<terminated> LogoutAutomationTest [Java Application] C:\Program Files\Java  
Successfully logged out.
```

2. Automated testing [Edit a subject] Eclipse :

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class UpdateTableCell {
    public static void main(String[] args) {
        // Set the path to your EdgeDriver executable
        // System.setProperty("webdriver.edge.driver", "path_to_your_edgedriver");

        // Initialize the EdgeDriver
        WebDriver driver = new EdgeDriver();

        try {
            // Step 1: Open the main page
            driver.get("http://localhost/studentrecordms/manage-subjects.php");

            // Step 2: Click on the "Subjects" menu item
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(50));
            WebElement subjectsMenu = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[contains(text(),'Subject')]")));
            subjectsMenu.click();

            // Step 3: Click on the "View" button in the dropdown
            WebElement viewButton = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[contains(@href, 'manage-subjects.php') and text()='View']")));
            viewButton.click();

            // Step 4: Click the "Edit" button for the specific subject
            WebElement editButton = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[contains(@href, 'edit-subject.php') and text()='Edit']")));
            editButton.click();

            // Step 5: Wait for the input field to load, clear its value, and enter a new one
            WebElement subjectInput = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//input[@name='sub4']"))); // Locate the input by its name
            subjectInput.clear(); // Clear the existing value
        }
    }
}

```

```
subjectInput.sendKeys("Kotlin"); // Enter the new value
```

```
// Step 6: Locate and click the "Update" button
```

```
WebElement updateButton = wait.until(ExpectedConditions.elementToBeClickable(
    By.xpath("//input[@type='submit' and @name='submit' and @value='Update Course']")));
updateButton.click();
```

```
// Confirmation
```

```
System.out.println("Subject updated successfully to 'Kotlin'.");
```

```
} catch (Exception e) {
    e.printStackTrace(); // Or log the exception
```

```
} finally {
    // Close the browser
    driver.quit();
```

```
}
```

View Course

Show 10 entries

Search:

S No	Course	Subject1	Subject2	Subject3	Subject4	Action
1	BACHELOR OF TECHNOLOGY	C LANGUAGE	OPERATING SYSTEM	DATA STRUCTURE	KOTLIN	Edit Delete
2	BACHELOR OF ARTS	ENGLISH	SOCIOLOGY	PHILOSOPHY	HISTORY	Edit Delete
3	BACHELOR OF COMPUTER APPLICATION	C LANGUAGE	SOFTWARE ENGINEERING	DISCRETE MATHEMATICS	COMPUTER NETWORKS	Edit Delete

		subid	cshort	cfull	sub1	sub2	sub3	sub4	dt_created	update_date
☐	Edit Copy Delete	1	3	Bachelor of Technology	C Language	Operating System	Data Structure	Kotlin	2024-02-15 17:23:31	28-12-2024
☐	Edit Copy Delete	2	5	Bachelor of Arts	English	Sociology	Philosophy	History	2024-02-16 11:43:21	NULL
☐	Edit Copy Delete	4	4	Bachelor of Computer Application	C Language	Software Engineering	Discrete Mathematics	Computer Networks	2024-02-16 22:27:14	NULL
☐	Edit Copy Delete	5	9	Master of Arts	History	Geography	Sociology	English	2024-02-16 07:24:15	28-12-2024

```
<terminated> UpdateTableCell [Java Application] C:\Program Files\Java
Subject edited successfully to 'Kotlin'.
```

3. Automated testing [Update a session] Eclipse :

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class UpdateSession {
    public static void main(String[] args) {
        // Set the path to your EdgeDriver executable
        // System.setProperty("webdriver.edge.driver", "path_to_your_edgedriver");

        // Initialize the EdgeDriver
        WebDriver driver = new EdgeDriver();

        try {
            // Step 1: Open the main page
            driver.get("http://localhost/studentrecordms/session.php");

            // Step 2: Click on the "Session" button
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(50));
            WebElement sessionMenu = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[contains(text(),'Session')]")));
            sessionMenu.click();

            // Step 3: Select the "2022-2023" radio button
            WebElement sessionRadioButton = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//input[@type='radio' and @value='2022-2023']"))); // Adjust the value if
needed
            sessionRadioButton.click();

            // Step 4: Click the "Update Session" button
            WebElement updateButton = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//input[@type='submit' and @value='Update Session']")));
            updateButton.click();

            // Confirmation
            System.out.println("Session updated successfully to '2022-2023'.");

        } catch (Exception e) {
            e.printStackTrace(); // Log any exceptions
        }
    }
}

```

```

    } finally {
        // Close the browser
        driver.quit();
    }
}
}

```

View Session

Session*

Current Session: 2022-2023

- ☐ 2017-2018
☐ 2018-2019
☐ 2019-2020
☐ 2020-2021
☐ 2021-2022
☒ 2022-2023
☐ 2023-2024
☐ 2024-2025
☐ 2025-2026
☐ 2026-2027

Update Session

				id	session	postingdate	status
☐				8	2017-2018	2024-02-15	0
☐				9	2018-2019	2024-02-15	0
☐				10	2019-2020	2024-02-15	0
☐				11	2020-2021	2024-02-15	0
☐				12	2021-2022	2024-02-15	0
☐				13	2022-2023	2024-02-15	1
☐				14	2023-2024	2024-02-15	0
☐				15	2024-2025	2024-02-15	0
☐				16	2025-2026	2024-02-15	0
☐				17	2026-2027	2024-02-15	0

```

<terminated> UpdateSession [Java Application] C:\Program Files\Java\jdk-
Session updated successfully to '2022-2023'.

```

3.Automation testing[Delete a student] eclipse :

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;
public class DeleteStudent {
    public static void main(String[] args) {
        // Set the path to your EdgeDriver executable
        // System.setProperty("webdriver.edge.driver", "path_to_your_edgedriver");

        // Initialize the EdgeDriver
        WebDriver driver = new EdgeDriver();

        try {
            // Step 1: Open the main page
            driver.get("http://localhost/studentrecordms/manage-students.php");

            // Step 2: Click on the "View Students" link
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(50));
            WebElement viewStudentsLink = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[contains(@href, 'manage-students.php') and contains(text(), 'View
Students')]"))));
            viewStudentsLink.click();

            // Step 3: Click on the "Delete" button for a specific student
            WebElement deleteButton = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[contains(@href, 'manage-students.php?del=1') and contains(@class,
'btn-danger')]"))));

            // Confirm the deletion action
            deleteButton.click();
            driver.switchTo().alert().accept(); // Handle the confirmation alert if it appears

            // Confirmation
            System.out.println("Student with ID 1 deleted successfully.");

        } catch (Exception e) {
            e.printStackTrace(); // Log any exceptions
        }
    }
}

```

```

    } finally {
        // Close the browser
        driver.quit();
    }
}
}
}

```

WELCOME ADMIN

View Students													
Show	10	entries	Search:										
SNo	RegNo	Name	Email	MobNO	Course	Subject	Action						
1	3485082443	RAHUL KUMAR SINGH	RAHUL@GMAIL.COM	1231231234	BCA	C LANGUAGE+SOFTWARE ENGINEERING+ DISCRETE MATHEMATICS	Edit	Delete					
2	6214384819	GARIM SINGH	GR@GMAIL.COM	4525631456	MA	HISTORY+GEOGRAPHY + SOCIOLOGY	Edit	Delete					
3	9820199657	LAKSHMI PRASANTH PILLAI	LAKSHMIP22CS@STUDENT.MES.AC.IN	8850447730	BSC CS	SOFTWARE TESTING AND QUALITY ASSURANCE+CLOUD COMPUTING+ MACHINE LEARNING	Edit	Delete					
4	7101818287	LAXMI PRASANT PILLAI	AEFKD@GMAIL.COM	1234567810	MCA	CONNECTED SUCCESSFULLYSOFTWARE TESTING AND QUALITY ASSURANCE+CLOUD COMPUTING+ MACHINE LEARNING	Edit	Delete					
5	2788973570	LAXMI PRASANT PILLAI	AEFKD@GMAIL.COM	1234567810	MCA	CONNECTED SUCCESSFULLYSOFTWARE TESTING AND QUALITY ASSURANCE+CLOUD COMPUTING+ MACHINE LEARNING	Edit	Delete					

idcoursesubjectfname mname lnamegendergnameocpincomecategorypchnational															
☐	 Edit	 Copy	 Delete	34	C Language+Software Engineering+ Discrete Mathemat...	Rahul	Kumar	Singh	Male	Balved Singh	Pvt. Job.	700000	general	no	Indian
☐	 Edit	 Copy	 Delete	59	History+Geography + Sociology	Garim		Singh	female	Ram Singh	Govt Job	700000	obc	no	Indian
Software testing and															

Software testing and

```

inated> DeleteStudent [Java Application] C:\Program Fi
Student with ID 1 deleted successfully.

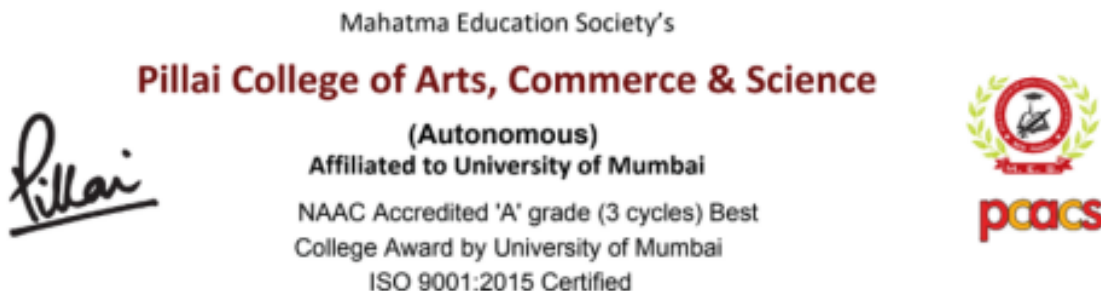
```

Conclusion :

The comprehensive testing strategy employed for the Student Record System ensures that the application meets the highest standards of functionality, performance, and reliability. By leveraging both manual and automated testing methodologies, critical functionalities such as course and student management, profile updates, and session handling have been rigorously validated.

Manual testing provided a detailed and systematic approach to validating individual features and edge cases, while automated testing enhanced efficiency by expediting repetitive tasks and increasing overall test coverage. The combination of these methodologies ensures that the Student Record System is robust, secure, and capable of delivering a seamless experience for all users.

This structured testing process not only identified and resolved potential issues but also built confidence in the application's ability to perform reliably in real-world scenarios. Moving forward, the framework established through this testing effort provides a solid foundation for ongoing maintenance, scalability, and future enhancements to the system.



CERTIFICATE

This is to certify that Miss. **Lakshmi Prasanth Pillai** of **T.Y B.Sc. C.S. Semester VI** has completed the project work of the Subject Of **Software Testing and Quality Assurance** during the academic year 2024-25 under the guidance of **Prof. Jabeen Masum Ma'am** is the partial requirement for the fulfillment of the curriculum of the Degree of Bachelor of Science in Computer Science, University of Mumbai.

Place:

Date:

Name & Signature of faculty

Name & Signature of external

Name & Signature of Co-ordinator

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